

Table 1. Statistical analyses of sex-specific hybrid effects on immune-predicted weight loss
parasiteLoad framework testing for deviations from additive expectations between subspecies

Dataset	Hypothesis Test	Analysis	p-value
Sex-Specific Effects			
Complete Dataset (n=304)	H0: Overall hybrid effect¹	Sex-specific effects	0.01742*
Complete Dataset (n=304)	H2: Females only	Sex-specific effects	0.18930
Complete Dataset (n=304)	H2: Males only	Sex-specific effects	0.03808*
Complete Dataset (n=304)	H3 vs H2²	Sex-specific effects	0.02668*
Constitutive Costs			
Uninfected Only (n=171)	H0: Overall hybrid effect ¹	Constitutive costs	0.54460
Uninfected Only (n=171)	H2: Females only	Constitutive costs	0.90810
Uninfected Only (n=171)	H2: Males only	Constitutive costs	0.43150
Uninfected Only (n=171)	H3 vs H2 ²	Constitutive costs	0.17300
Infection Dominance			
By Infection Status (n=304)	H0: Overall hybrid effect ¹	Infection dominance	0.19800
By Infection Status (n=304)	H3 vs H2 ²	Infection dominance	0.22850
Males Only - Infection (n=153)	H0: Overall hybrid effect ¹	Males infection-specific	0.21110
Males Only - Infection (n=153)	H3 vs H2 ²	Males infection-specific	0.11620
Infected Only (n=133)	H0: Overall hybrid effect ¹	Infected mice only	0.29290
Infected Only (n=133)	H2: Females only	Infected mice only	0.47360
Infected Only (n=133)	H2: Males only	Infected mice only	0.48330
Infected Only (n=133)	H3 vs H2²	Infected mice only	0.02746*

¹ Tests whether hybrids show overall deviation from linear expectations

² Critical test for sex-specific effects (H3 vs H2 comparison)

Significance codes: ***p < 0.001, **p < 0.01, *p < 0.05, †p < 0.1

Positive α indicates hybrid resistance (reduced costs); negative α indicates susceptibility (increased costs)